

Using Physics-Informed Neural Networks to Solve Differential Equations

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In this project, we solved various Differential Equations using Physics Informed Neural Networks (PINNs). PINNs use laws of Physics, which are mostly governed by differential equations. It incorporates these equations in the loss function for training our Neural Networks

For this last submission, I have selected the Poisson Equation to solve using PINNs. The Poisson Equation is used in various fields, such as Electrostatics and Fluid Dynamics.

Poisson's Equation can be written as:

$$\nabla^2 u = f(x)$$

Our problem is given as:

$$\nabla^2 u = -\pi^2 \sin(\pi x)$$

The boundary conditions are:

$$u(-1) = 0, u(1) = 0$$

Let $g(x)$ be a function to describe these boundary conditions

$$g(x) = 1 - |x|$$

The PDE can be written as:

$$\frac{\partial^2 u}{\partial x^2} = -\pi^2 \sin(\pi x)$$

The exact solution of this PDE is :

$$u(x) = \sin(\pi x)$$

So the residual can be written as:

$$\frac{\partial^2 u}{\partial x^2} + \pi^2 \sin(\pi x) = 0$$

Procedure Used:

- We defined the equation and its boundary conditions.
- Then constructed a fully connected Neural Network to approximately estimate the solution.
- Designed our loss function according to the PDE as,
 - (a) The error in satisfying the boundary conditions.
 - (b) The error in satisfying the Poisson equation over a set of collocation points.

Neural Network Architecture:

We used a fully connected feedforward network. The input to the network is the spatial coordinate x , and the output is the value of the function $u(x)$. The network has the following structure:

- Input layer: 1 node takes input x
- Hidden layers: 5 hidden layers with 50, 50, 20, 50 and 50 neurons, respectively
- Activation function: tanh
- Output Layer: 1 node provides output $u(x)$

The loss function is defined as:

$$L_{\text{total}} = L_{\text{BC}} + L_{\text{PDE}}$$

$$L_{\text{BC}} = \frac{1}{2} \sum_{i=1}^{N_u} (u(x_i) - u_{\text{BC}}(x_i))^2$$

$$L_{\text{PDE}} = \sum_{i=1}^{N_f} \left(\frac{\partial^2 u}{\partial x^2} - f(x) \right)^2$$

N_u and N_f are boundary and Collocation points respectively

Data Generation:

The analytical solution over the full range of our domain is used

$$x \in [-1, 1]$$

A total of 500 points are divided into boundary points and collocation points. The exact solution of the Poisson equation is

$$u(x) = \sin(\pi x)$$

This is used to compare the results of PINN.

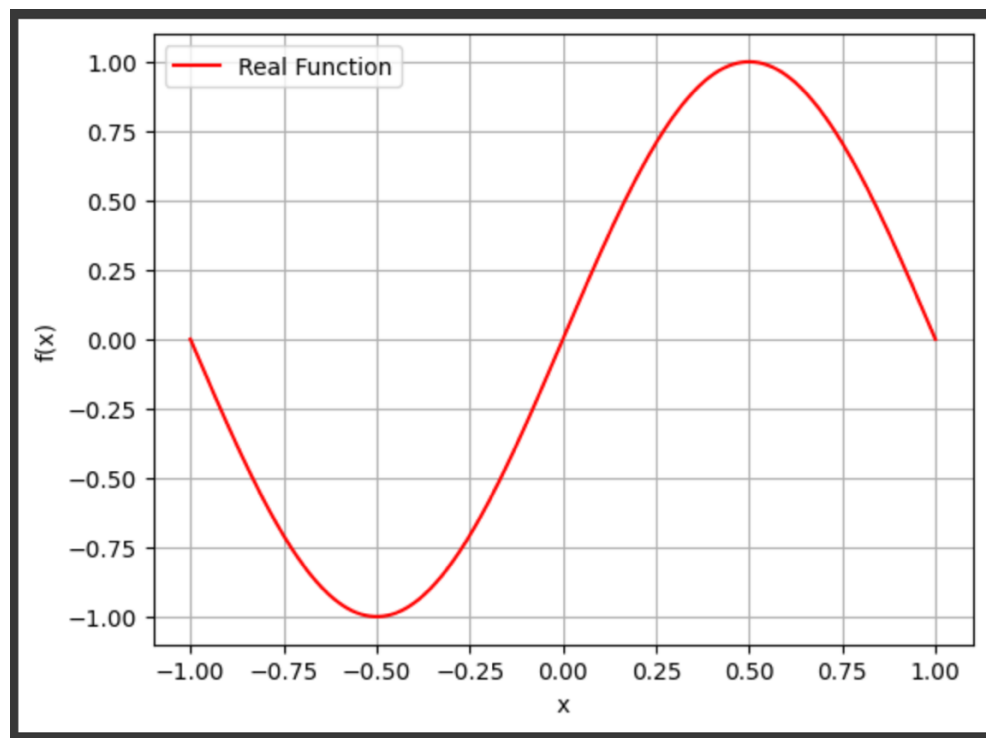
Training Process uses

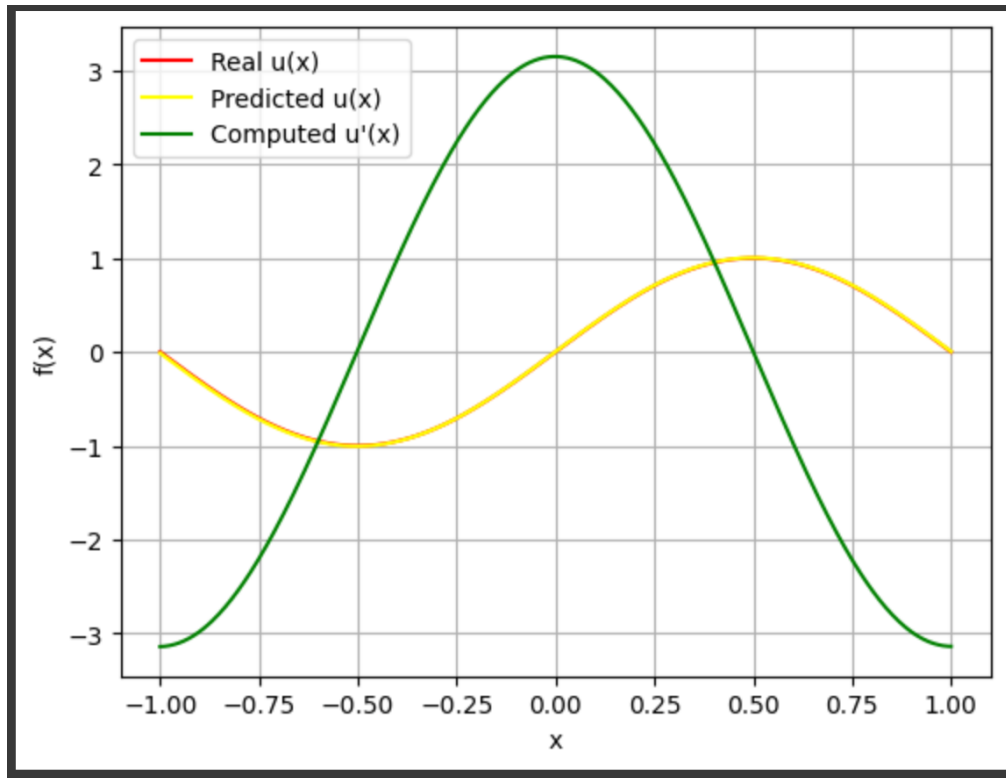
- Adam optimizer (learning rate = 10^{-3})
- Total 5000 training steps

The loss function is minimised by adjusting the weights, and this is done by using Backpropagation.

The network learns to approximate the solution and its derivatives ($u'(x)$).

Results:





As we can see from the plots produced, the predicted solution almost matches the real solution. The final error between the predicted and original solution is in the order of 10^{-6} .

This project showed the application of PINNs to solve differential equations. In this particular case, we solved the Poisson's equation. By incorporating the physical laws (governed by differential equations) into loss functions, this network gave us the approximate solution with high accuracy.